

Speaker Script: parts_of_a_deployment.pptx

Duration: 40 Minutes | Slides: 13 | Role: DOE / ORISE Marine Energy Fellow

THE PRESENTATION SPINE

Anchor	Core Focus	Slides	Time
DRAW	What it costs	3, 4	03:30
DIVIDE	Who gets it	5	10:30
PREDICT	What's coming	6, 7	15:00
PROTECT	What can't break	8	24:00
PROVE	What really happened	9, 10	28:30
ADMIT	Five corrections	11	34:30
ASK	Open questions / Closing	12, 13	36:30

Slide 1 • Title • Parts of a Deployment

Anchor: Open • 45 sec • 0:00

Stage Direction: Establish a steady, conversational pace. Do not linger on this slide. Move the room forward rapidly.

A year on one question: how do you run a scientific instrument on a battery that only refills when the sun cooperates?

I'm Aditi. I've been the DOE and ORISE marine energy fellow here for the past year. My background is thermal and renewable energy systems, so I came at this as an energy budget problem and learned the machine learning on the way.

Forty minutes, thirteen slides, and I'd like the hard questions at the end.

Slide 2 • Agenda • This year, laid out like a planet

Anchor: Open • 2 min 45 • 0:45

Stage Direction: Walk through the five bars on the slide from bottom to top. Take one clear breath for each bar without elaborating yet.

Each layer sits on the one below it, which is genuinely how it got built. I want to show the year as a cross-section, because that's honestly the order it happened in. Each month sat on top of the month before.

Core is power measurement, what the thing draws. Inner mantle is dividing that power. Outer mantle is forecasting, guessing tomorrow. Crust is a safety floor that overrules both. And the surface is the real deployment, where the ocean gets a vote.

The shape of the story is in that stack. A wrong number at the core invalidated work three layers up, which is the part I didn't expect when I started.

One flag before I go. There's a slide near the end of things I got wrong. I'm not saving them for a confession, they just land better once you've seen the work they affected.

Slide 3 • Core • What does this system actually draw?

Anchor: Draw • 4 min 30 • 3:30

Stage Direction: Point to the left card as you begin reading the baseline metrics checklist.

Four measured numbers on IMTA, all confirmed. One of them had the wrong label on it for most of the year.

Context for anyone who hasn't seen this system. IMTA is our deployed buoy, integrated multi-trophic aquaculture, off the Gulf Coast of Alabama. Panel on top, battery in the hull, camera on a tether underwater, cell link so we can talk to it.

Here are the baseline metrics: a 100 W panel, a 1200 Wh battery, 1.7 W standby draw, and 33.7 W awake draw. These four numbers recur on every slide after this.

1.7 watts is standby, and that's the particle board only. The little cell link that never really switches off, so we can wake the rest remotely. Published range for that board is half a watt to two and a half, so 1.7 sits right where it should.

33.7 watts is the whole package awake, Jetson and cameras together. Not the camera on its own. I had that labelled wrong until about two weeks ago, which is the red box at the bottom.

Stage Direction: Direct attention to the right card for AMEC, indicating that the open item is intentional.

AMEC is the other system, still being sized rather than deployed. We know the draw, about 90 watts active phase once the marine instruments are duty-cycle adjusted. What we don't have is the battery and panel sizing to match it.

[IF ASKED / FILTER]

Q: I'd have expected the Jetson to be higher than that.

A: You're right, and that's exactly the correction. A Jetson alone is around 15 W. The 1.7 is only the particle board with everything else powered down.

Q: Where did the measured numbers come from?

A: Telemetry from the deployment, cross-checked with Corey on the hardware side. Emily has been my route into the IMTA system data.

Q: Is AMEC blocked?

A: On measurement, not on method. The sizing math is worked out, it needs real numbers off a real install to validate against.

Slide 4 • Core chart • Same panel, same system. Only the site changes.

Anchor: Draw • 2 min 30 • 8:00

Stage Direction: Explicitly reference what is fixed before highlighting what varies to keep the audience grounded.

Hold the hardware fixed, change only the location, and uptime swings by a factor of three.

Two things held fixed: a 100 watt panel, and this exact system, 1.7 standby and 33.7 awake. The only thing changing is where you put it.

What comes out is hours per day the payload can stay awake. 14.4 hours in Hawaii. 5.1 in Anchorage. Same hardware, nearly three times the uptime, purely from latitude and weather.

Flip it for procurement, which is the more useful direction: to buy Anchorage back up to Hawaii's uptime, you fit roughly a 150 watt panel instead of a 100.

[IF ASKED / FILTER]

Q: What is a peak sun hour?

A: Not daylight hours. The equivalent number of hours at full rated sun, so it already folds in angle, season and typical cloud.

Q: Where does the 0.85 come from?

A: Standard derate for wiring, controller efficiency and panel temperature. A convention, not something I measured.

Slide 5 • Inner mantle • Given those numbers, how do you divide the power?

Anchor: Divide • 4 min 30 • 10:30

Stage Direction: Narrate through the five printed steps sequentially down the slide, rather than performing active calculations.

Once you trust the draw, scheduling is arithmetic. Pay standby first, spend what's left.

This runs once every morning and answers one question: how much of today can the payload afford to be awake.

Step one, read the battery and tomorrow's forecast. Say 600 watt hours in the pack and 9.6 watts average forecast.

Step two, does the forecast clear standby plus a safety margin. Standby is 1.7, margin is 5, so the bar is 6.7. We're at 9.6, so yes.

If it were no, the answer is simple and boring: stay in standby, payload duty zero, ride the battery down and wait for better weather.

Because it's yes, we allocate by priority, and standby gets paid first, always. That leaves 7.9 watts. Divide by the 33.7 the package needs and you get 23 percent of the day, about five and a half hours awake.

Stage Direction: Draw focus to the right card highlighting the adjustment knob to close the slide on a talking point.

The 5 watt margin is the tuning knob. Raise it and the system gets conservative and wastes sun. Lower it and you get more data and more risk. Right now it's set by judgment, not derived from anything.

[IF ASKED / FILTER]

Q: What if the forecast is wrong?

A: That's the next layer, and the layer after that is what catches it when the forecast and the scheduler are both wrong.

Q: Why daily and not hourly?

A: The daily solar cycle is the dominant signal and hourly re-planning adds churn. Worth testing, not ruled out.

Q: Does it generalise past two loads?

A: The cascade does, you keep spending the remaining watts in priority order. It hasn't been tested with sonar and UV-C competing, which is one of my open questions.

Slide 6 • Outer mantle • Can the schedule see tomorrow coming?

Anchor: Predict • 6 min • 15:00

Stage Direction: This is your longest slide. Treat Acts 1, 2, and 3 as distinct, separate beats. Take a breath between them.

Three acts, and I was wrong in two of them. That is the useful part.

This is the layer where I learned the most, almost entirely by being wrong. I'll tell it in the order it actually happened, not the tidy version.

Act one. I built a benchmark, held out the last 48 hours, and the dumbest possible model won. Naive seasonal, which literally repeats yesterday, came out near perfect. That felt suspicious, and it was. It was leakage, the 24 hour lag features it was scored on still reached back into the training window. For a while I believed ML barely mattered here, and that belief was built on a broken test.

Act two. Rebuilt with a strict 28 day train, 7 day test split, eleven models. The ranking flipped completely. LSTM wins at 0.93 watts average error, naive falls to the bottom where it belongs.

Act three. This one I only understood in the last few weeks. I wasn't forecasting sunlight. I was forecasting panel watts, the power actually arriving at the battery. And once the battery is full, the controller caps that regardless of how bright it is outside. So the model was closer to a load forecast than a solar forecast.

Stage Direction: Turn to the right-hand card. Drop your volume slightly and slow right down for the core takeaway.

Eight classical models landed within 0.95 to 1.23 watts of each other. The spread between completely different model families was smaller than the spread one change to the data split opened up. Model choice barely moved the number. Data composition moved it a lot.

And act three reframes the question rather than just invalidating it. There are two things worth predicting here, what the sun delivers and what the battery holds, and they can't be cleanly separated. Knowing tomorrow's sun without knowing tonight's charge doesn't tell you what the system can afford to run. Paul actually raised this during the dry run, and he's entirely right: you need both.

[IF ASKED / FILTER]

Q: What is MAE?

A: Mean absolute error. On average, how many watts off the prediction is. Same units as the thing predicted, which is why I use it.

Q: Why LSTM over Prophet?

A: Prophet assumes a repeating seasonal shape and smooths. On 35 days at sea the 24 hour lookback caught the intra-day swing better. Prophet was 3.58 W, about four times worse.

Q: What would you do differently?

A: Jack Mead's checklist, which isn't on this slide: problem before model, data is ninety percent of the job, don't pile on variables, source external data properly, clean before training. My leakage was a data problem, not a model problem.

Slide 7 · Benchmark chart · The verified 11-model benchmark

Anchor: Predict · 3 min · 21:00

Stage Direction: Strict ban on the words 'log scale.' Focus on the extreme visual heights on a linear axis.

Same data, same models. The only thing that changed was the split, and the ranking flipped.

This is a linear axis. Naive towers over everything, which makes the point on its own.

Four distinct groups: LSTM at 0.93 watts, the classical models clustered just behind it, Prophet at 3.58, and the naive baselines out at 19.3.

For scale, 0.93 watts against an 85 watt midday peak is about one percent. If you plotted predicted against actual, the gap would be thinner than the line itself.

Twenty eight days to train, seven unseen days to grade on. That split is the only thing that changed between the broken benchmark and this one.

[If room is quiet]: There's a live version of this scheduler I can show after, if it's useful.

Slide 8 • Crust • What actually stops the battery from dying?

Anchor: Protect • 4 min 30 • 24:00

Stage Direction: Emphasize that this is software logic, NOT hardware circuitry. Clarity is key.

It isn't hardware. It's a rule in the scheduler that overrules the forecast, and across 41 days it never had to be broken.

Worth being precise, because the name misleads. This is not hardware. There's no protection circuit doing this. It's a rule living inside the scheduler.

Before committing to any duty cycle from the layer below, it simulates the battery forward hour by hour and refuses to output a schedule that would cross the floor. It does not care what the forecast said. The forecast can be wrong and the floor still holds.

Battery next hour equals battery now, plus solar in, minus standby, minus duty times package. That's the whole thing.

And the result: across all 41 days of telemetry, no schedule ever crossed the floor.

Stage Direction: Point to the chart on the right. Give it a silent 2-second beat before speaking.

This is real data, not the simulation. Battery voltage sits on a flat plateau near float almost the entire deployment, even through stretches with no solar input. That plateau is the floor holding. The two sharp drops are exactly the events it exists to catch.

[IF ASKED / FILTER]

Q: Where does the floor number come from?

A: 240 Wh, 20 percent of a 1200 Wh LiFePO4 pack. A chemistry convention for safe depth of discharge, not

something I derived from these cells.

Q: Has it ever actually refused a schedule?

A: In replay across the 41 days it never had to. That's a mild result and I'd say so: conditions were never bad enough to test it properly.

Q: Is this a digital twin?

A: It's the honest version of one. State of charge only. A real twin adds thermal and degradation models and runs live on the buoy instead of in replay.

Slide 9 · Surface, Gulf Coast · One dip, two completely different causes

Anchor: Prove · 3 min 30 · 28:30

Stage Direction: Build up to the mid-slide revelation intentionally, then pause entirely to let the twist land.

What looked like a dying panel was a healthy controller doing its job, with a real eight day outage hiding inside it.

Then the real world gets a vote. On the Gulf Coast buoy, harvest looked like it was falling off a cliff and the first read was roughly 60 percent panel degradation. Fouling, wear, something physical. That read held for weeks.

It was wrong. Most of it was curtailment, the controller deliberately backing off because the battery was already full. Confirmed against the controller's own state codes, cycling bulk, absorption, float, every day. The panel was fine.

[Pause here for 3 full seconds. Let the reversal settle.]

But sitting inside that same window was a genuine 8 day outage, June 25 to July 2. State pinned at Off for a hundred percent of those minutes, never charging, and never logging a fault code.

One explanation had been covering for the other. The curtailment story was true, and it was hiding a second thing that wasn't.

[IF ASKED / FILTER]

Q: What caused the outage?

A: Still open. Telemetry alone can't tell me. It's one of the questions I'd hand over.

Q: How did you catch the curtailment?

A: Corey pointed at the charge controller state codes. Once you read state rather than just harvest, it's unambiguous.

Slide 10 · Surface, Hawaii · Second site, different hardware, same method

Anchor: Prove · 2 min 30 · 32:00

Stage Direction: Deliver the core contrast clearly—data anomalies vs. local ground human logs. Credit Paul explicitly.

The method transferred to hardware I had never touched, and then a person told me the thing the data could not.

Second site, Hawaii Island, a Blue Robotics install. Different hardware, different owner, I've never touched it. Daily harvest sat flat at 180 to 240 watt hours while cloud cover swung from 26 percent to 93.

Harvest decoupled from weather is the curtailment signature. It reproduced on a system I had no hand in building, which is the only reason I trust the method at all.

Then a two day collapse that lined up exactly with a wind spike. That turned out to be the panel physically tearing off its mount. Tony confirmed it.

I could see the anomaly. I could not have told you the cause. Someone who could see the roof told me.

Paul made exactly this point in the dry run: the hard part of remote sensing is knowing what happened on the ground, and that depends on logs and communication more than on analysis. Crediting him is both true and generous.

Slide 11 • Corrections • Five things I had wrong

Anchor: Admit • 2 min • 34:30

Stage Direction: Sprint mode. Read these lines rapidly like a rapid-fire ledger. No re-explaining allowed.

Four of the five were caught by someone who could see the hardware. Not by better modelling.

All five are on one slide here. Go fast, one line each, do not re-explain. You've seen four of them already.

- 33.7 watts was labelled the camera. It's the whole package awake.
- 1.7 watts was labelled Jetson plus router. It's the particle board alone.
- Panel degradation was curtailment, with a real outage hidden inside it.
- The naive model winning was leakage in my train test split.
- The solar forecast was not forecasting solar.

[Hard brake. Slow down completely for these last two sentences.]

Four of those five were caught by a person who could see the hardware, not by better analysis. The fifth I caught by re-checking my own split.

So the thing I'd tell whoever picks this up: remote diagnosis gets you a short list of candidates, not an answer. The last step is always a person who was there.

Slide 12 · Research · One open question per layer

Anchor: Ask · 2 min · 36:30

Stage Direction: Do not read all six rows. Pick exactly two, name them, and then pivot straight to the prompt at the bottom.

If you had to fund one of these, which one?

Rather than everything undone, one question per layer, the one I'd actually chase.

The big one is the outer mantle: predict solar and battery state together, not solar alone. Bring in a ground truth irradiance signal decoupled from the battery, use battery state as the second input, and test whether the joint model schedules better than either half alone.

And at the surface, both Hawaii hardware questions are closed now. The next one is whether curtailment versus fault can become predictive, flagging a likely fault before it fully develops rather than explaining it afterwards.

[Stop talking completely after this next question. Let the silence work.]

If you had to fund one of these, which one?

Slide 13 · Closing · The surface kept correcting the core

Anchor: Close · 1 min 30 · 38:30

Stage Direction: Bring the narrative loop to a definitive, clean stop. Thank your team by name.

Every time I went out to the real deployment, it sent me back down to fix something I thought was settled.

The shape of the year in one sentence, which is on the slide. A core fact, a schedule built on it, a forecast layered on the schedule, a floor protecting all of it, and a surface where the real world gets the final word.

And the honest version: the surface kept correcting the core. That's the part I didn't expect, and the part I'll carry.

Thanks to MarineSitu and ORISE for the year. Paul for the guidance and for treating your questions as worth taking seriously. Mitchell for the technical framing. Corey and Tony for telling you what the hardware was doing when the telemetry couldn't.

Then hand it over: questions, and I'd genuinely like the hard ones.

[Optional Backup Track]: I also have a live version of the scheduler in the browser, happy to run someone's numbers through it.

